

CBCS SCHEME

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BCS503

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Theory of Computation

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C														
Q.1	a.	Define the following with example : i) Language ii) String iii) Power of an alphabet.	3	L1	CO1															
	b.	Define DFA. Draw a DFA to accepts. i) The set of all strings that contain a substring aba. ii) To accept the strings of a's and b's that contain not more than three b's. iii) $L = \{w \in \{a, b\}^* : \text{No 2 consecutive characters are same in } w\}$.	10	L3	CO1															
	c.	Convert the following NFA to DFA. → <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>p</td><td>{p, q}</td><td>{p}</td></tr><tr><td>q</td><td>{r}</td><td>{r}</td></tr><tr><td>r</td><td>{s}</td><td>ϕ</td></tr><tr><td>* s</td><td>{s}</td><td>{s}</td></tr></table>		0	1	p	{p, q}	{p}	q	{r}	{r}	r	{s}	ϕ	* s	{s}	{s}	7	L2	CO1
	0	1																		
p	{p, q}	{p}																		
q	{r}	{r}																		
r	{s}	ϕ																		
* s	{s}	{s}																		
OR																				
Q.2	a.	Define the following with example : i) Alphabet ii) Reversal of string iii) Concatenation of Languages.	3	L1	CO1															
	b.	Design a DFA for the Language : $L = \{w \in \{0, 1\}^* : w \text{ is a string divisible by } 5\}$.	7	L3	CO1															
	c.	Define NFA. Obtain an ε - NFA which accepts strings consisting of 0 or more a's , followed by 0 or more b's followed by 0 or more C's. Also convert it to DFA.	10	L2	CO1															
Module – 2																				
Q.3	a.	Define Regular expression. Write the regular expression for the following languages : i) Strings of a's and b's starting with a and ending with b. ii) Set of strings that consists of alternating 0's and 1's. iii) $L = \{a^n b^m, (n + m) \text{ is even}\}$. iv) $L = \{w : w \bmod 3 = 0, \text{ where } w \in \{a, b\}^*\}$.	10	L2	CO2															

	b.	Minimize the following finite automata using Table filling algorithm : <table><tr><th>δ</th><th>a</th><th>b</th></tr><tr><td>$\rightarrow$ A</td><td>B</td><td>A</td></tr><tr><td>B</td><td>A</td><td>C</td></tr><tr><td>C</td><td>D</td><td>B</td></tr><tr><td>*</td><td>D</td><td>A</td></tr><tr><td>E</td><td>D</td><td>F</td></tr><tr><td>F</td><td>G</td><td>E</td></tr><tr><td>G</td><td>F</td><td>G</td></tr><tr><td>H</td><td>G</td><td>D</td></tr></table>	δ	a	b	$\rightarrow$ A	B	A	B	A	C	C	D	B	*	D	A	E	D	F	F	G	E	G	F	G	H	G	D	10	L2	CO2
δ	a	b																														
$\rightarrow$ A	B	A																														
B	A	C																														
C	D	B																														
*	D	A																														
E	D	F																														
F	G	E																														
G	F	G																														
H	G	D																														
OR																																
Q.4	a.	Construct ε - NFA for the following Regular expression : i) $(0 + 1) 0 1(1 + 0)$ ii) $1(0 + 1)^* 0$ iii) $(0 + 1)^* 0 1 1^*$	6	L1	CO2																											
	b.	Obtain the Regular expression that denotes the language accepted by Fig. Q4(b). Fig. Q4(b) Using Kleene's theorem.	6	L3	CO2																											
	c.	State the Pumping Lemma for the Regular Languages. And also prove that the following languages are not regular. i) $L = \{0^n 1^m \mid n \leq m\}$ ii) $L = \{0^n 1^m 2^n \mid n, m \geq 1\}$.	8	L1	CO2																											
Module – 3																																
Q.5	a.	Design CFG for the following languages : i) $L = \{a^n b^{n+3}, n \geq 0\}$ ii) $L = \{a^i b^j c^k, j = i + k, i \geq 0, k \geq 0\}$ iii) $L = \{w \mid w \bmod 3 > 0 \text{ where } w \in \{a\}^*\}$ iv) $L = \{a^m b^n \mid m \neq n\}$ v) Palindromes over 0 and 1.	10	L3	CO3																											
	b.	Consider the grammar G with productions. $S \rightarrow A b B / A / B$; $A \rightarrow aA / \varepsilon$; $B \rightarrow a B / b B / \varepsilon$. Obtain LMD , RMD and parse tree for the string aabab. Is the given grammar ambiguous?	10	L2	CO3																											
OR																																
Q.6	a.	Define the following with example : i) Context free grammar ii) Left most Derivation iii) Parse tree iv) Ambiguous grammar.	4	L1	CO3																											
	b.	Design PDA for the language : $L = \{a^i b^j c^k \mid i + k = j, i \geq 0, k \geq 0\}$ and show the moves made by the PDA for the string aabbcc.	10	L3	CO3																											

	c.	Convert the following CFG's to PDA : $S \rightarrow aA$; $A \rightarrow aABC / bB / a$; $B \rightarrow b$; $C \rightarrow c$.	6	L2	CO3
Module – 4					
Q.7	a.	Define CNF. Convert the following CFG to CNF $E \rightarrow E + T / T$ $T \rightarrow T * F / F$ $F \rightarrow (E) / I$ $I \rightarrow Ia / Ib / a / b$.	10	L2	CO4
	b.	Show that $L = \{0^n 1^n 2^n / n \geq 1\}$ is not context free.	4	L2	CO4
	c.	Prove that the family of context free languages is closed under union and concatenation.	6	L1	CO4
OR					
Q.8	a.	Define Greibach Normal Form. Convert the following CFG to GNF. $S \rightarrow AB$; $A \rightarrow aA / bB / b$; $B \rightarrow b$.	6	L2	CO4
	b.	Consider the following CFG : $S \rightarrow ABC / BaB$ $A \rightarrow aA / BaC / aaa$ $B \rightarrow bBb / a / D$ $C \rightarrow CA / AC$ $D \rightarrow \epsilon$ i) What are useless symbols? ii) Eliminate ϵ - productions , Unit productions and useless symbols from the grammar.	10	L3	CO4
	c.	Prove that the following languages are not context free. i) $L = \{a^i / i \text{ is prime}\}$ ii) $L = \{a^{n^2} / n \geq 1\}$.	4	L2	CO3
Module – 5					
Q.9	a.	Define a Turing machine and explain with neat diagram, the working of a basic Turing machine.	6	L1	CO4
	b.	Design a Turing machine to accept the language, $L = \{a^n b^n c^n / n \geq 1\}$. Draw the transition diagram and show the moves for the string aabbcc.	14	L4	CO4
OR					
Q.10	a.	Design a Turing machine to accept palindrome over $\{a, b\}$ and draw the transition diagram.	12	L4	CO5
	b.	Write a short notes on : i) Recursively Enumerable Language. ii) Multitape Turing Machine.	8	L1	CO5
